Definition. Suppose all the Riemann sums for a function $f(x)$ on an interval $[a$ get arbitrarily close to a single number when the lengths Δx_1, ..., Δx_n are ma small enough. Then this number is called the **integral** of $f(x)$ on $[a, b]$ and it denoted

$$\int_a^b f(x)\,dx.$$